

- (ii) Toon aan dat die horisontale afstand wat die reddingspak afgelê het is 1869 m.
Show that the horizontal distance covered by the parcel is 1869 m.

$$x - x_0 = v_{0x} \cdot t$$

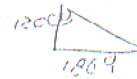
$$= 119,44 \cdot (15,65)$$

$$= 1869 \text{ m}$$

- (iii) Bepaal die hoek ϕ . / Determine the angle ϕ .

$$\frac{1869}{1200} = \tan \phi$$

$$\phi = 57,3^\circ$$



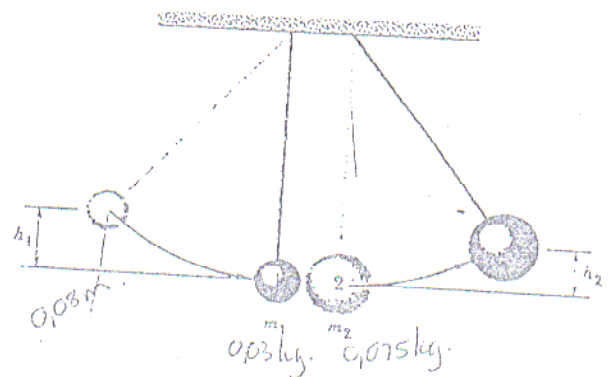
(2)

(1)

VRAAG / QUESTION 2

Twee metaal balle is vas aan toue soos in meegaande skets. Bal 1 (massa $m_1 = 30$ gram) word opgelig tot hoogte $h_1 = 8,0$ cm en dan gelos om elasties te bots met bal 2 met massa $m_2 = 75$ gram.

Two metal spheres are suspended by cords as shown in the figure. Sphere 1 (mass $m_1 = 30$ gram) is pulled to a height $h_1 = 8,0$ cm and then released. It undergoes an elastic collision with sphere 2 whose mass is $m_2 = 75$ gram.



- (a) Toon aan dat speed die v_{1i} van 1 net voor die botsing is $v_{1i} = 1,252$ m/s.

Show that the speed v_{1i} of 1 just before the collision is $v_{1i} = 1,252$ m/s.

$$U_{\text{voor}} + K_{\text{voor}} = U_{\text{na}} + K_{\text{na}}$$

$$mgh + 0 = 0 + \frac{1}{2}mv^2$$

$$gh = \frac{1}{2}v^2$$

$$\sqrt{2gh} = v$$

$$v = 1,252 \text{ m/s}$$

(3)

- (b) Na die botsing is $v_{1f} = -0,537$ m/s. Toon aan dat $v_{2f} = 0,7156$ m/s.
After the collision $v_{1f} = -0,537$ m/s. Show that $v_{2f} = 0,7156$ m/s.

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f} \quad (\text{Behoud van momentum})$$

$$(0,03)(1,252) + 0 = (0,03)(-0,537) + (0,075)(v_{2f})$$

$$0,03756 = -0,01611 + 0,075v_{2f}$$

$$\therefore v_{2f} = 0,7156 \text{ m/s}$$

Vraag / Question 2(c)(vervolg)(continue) / ...

(3)